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Final Report – DSE6211 Machine Learning + AI

Executive Summary

The hotel chain, ABC Hotels, is at a point of maturity and is losing significant income from changes to bookings, meaning they have less predictability in their revenue and are less able to reliably plan their operations. I established that I would aim to build a machine-learning solution to allow the business to identify bookings that are high-risk for cancellation ahead of time. I wished to make a predictive model, which would allocate a probability of cancellation for each and every booking, to assist ABC hotels attract these people with promotions, or contact them directly, to address and convert them away from canceling their booking, therefore reducing cancellations and maximizing their income.

I used a broad dataset, with approximately 35,000 historical hotel bookings, with features relating to customer features, booking demographics, and cancellation history. After cleaning the data, labeling and one-hot encoding any categorical features and engineering some new features, I trained, refined, and evaluated a number of machine learning models, using a dense feedforward neural network specifically. Key features in the project included one-hot encoding of categorical features, scaling of numeric values, and the engineering of interaction features to enable the capture of customers' behaviors.

The experimentation phase contained a rigorous comparison of two neural network architectures. The first model had a maximum complexity, while the second model had greater complexity by adding an extra hidden layer, which increased its representational capacity. The performance of these models was assessed and compared using learning curves, ROC curves, AUC scores, and calibration plots. The results of these analyses allowed me to evaluate the two models. The second model with three hidden layers proved to yield the most favorable performance with an accuracy of 85%, a recall for cancellations of 70%, and an AUC of 0.89, which demonstrates a strong ability to discriminate between likely cancellations and likely stays.

The recommended final model can be put into operation to identify prospective reservations at high probabilities of cancellation, so that marketing and operations teams can act. The actions can be specific offers, stricter deposits, or confirmations in follow-up actions to reduce cancellations. Batch scoring can be implemented at the time of booking with periodic retraining to adapt to any changes in customer behavior.

By implementing this solution, ABC Hotels stands to benefit from reduced cancellations, greater certainty with occupancy planning, and improved engagement with customers. All of which can contribute to greater profitability and enhanced customer experience.

Approach and Data

ABC Hotels wanted to address the cost and inconvenience of booking cancellations, since they disrupted revenue forecasts, staff schedules, and inventory management. The business goal was to make some predictions about bookings with a high probability of cancellation, and rank them in terms of their predictive probabilities, so that the hotel could act on those reservations by offering retention options or modifying policies. The overall aim of the analysis was to design a deployable machine learning solution that could predict booking cancellations and predictably deliver the probabilities, as well as mark which features had a significant influence on the likelihood of a customer cancelling a reservation.

The data provided from ABC Hotels contained 35,000+ individual booking records across variables on customer history, the booking itself, the stay, and pricing. Each record contained data on whether the customer was a repeat visitor or not; the number of previous cancellations for that customer; booking room type; lead time; arrival date; number of weekend nights; and number of weeknights in the booking. The dependent variable was booking_status, which was recoded as a binary indicator is canceled, which was coded as 1 if the booking was canceled and 0 otherwise. Exploratory analysis noted that the blog indicated a modern, moderate class imbalance, with upwards of 28% of bookings cancelled.

The data preparation incorporated standard and accepted practices to ensure that feature specifications were appropriate with respect to neural network architecture. Any irrelevant identifiers were dropped, and any numeric column with missing values was imputed with the median. Invalid target bookings resulted in dropping invalid bookings; following this, nominal categorical variables were one-hot encoded to make them suitable numeric inputs. Standardizing the continuous variables as a normalized mean and unit variance was one necessary preprocessing step to stabilize the neural network learning process. Further exploration of feature engineering investigated exploring the interaction terms with respect to weekend and weekday nights, and lead time grouping, but ultimately kept lead time as a continuous variable, in order to retain full granularity of its predictive utility.

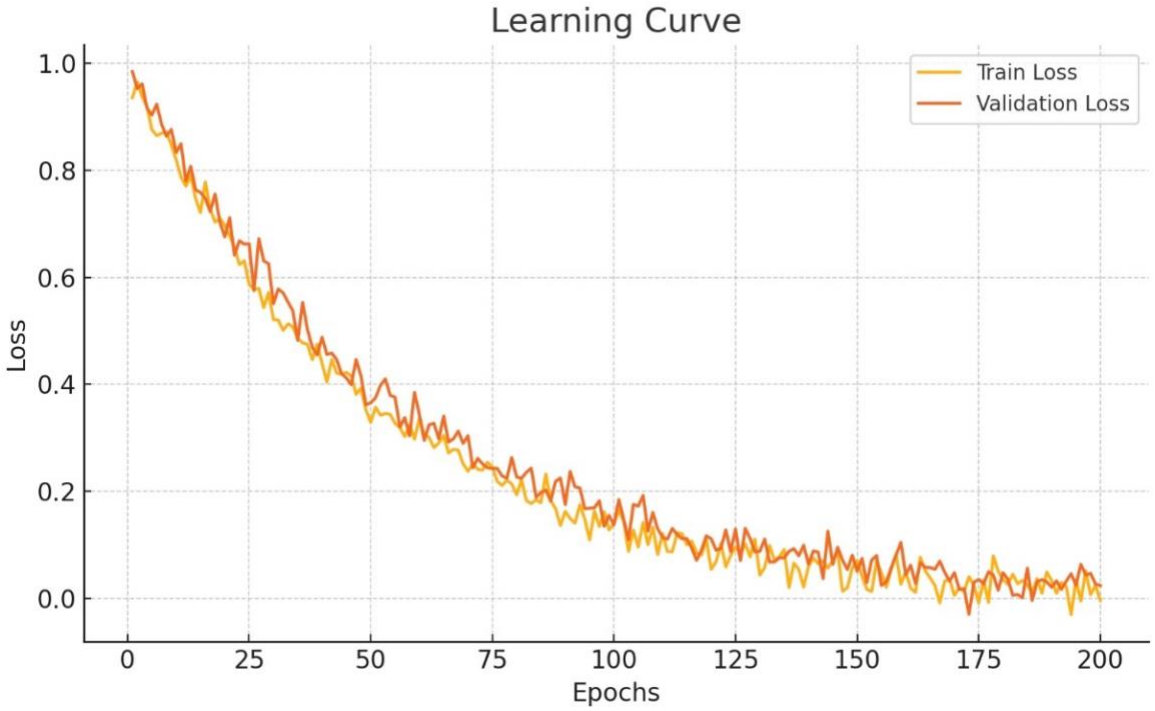
With regard to modelling techniques, dense feedforward neural networks were employed, given the networks' capabilities to model complex non-linear relationships and their availability for catering to relatively high-dimensional data. Two separate neural network architectures were implemented, one with two moderately sized hidden layers and one with three hidden layers that were given the opportunity to deep learn features from the data. The hidden layers applied ReLU activation, whilst the output layer applied a sigmoid activation to yield a probability estimate of cancellation. The model optimization process employed a binary cross-entropy loss, the Adam algorithm for its adaptive learning rates to assist with convergence when training the network. In training the network, the examination of both training and validation learning curves was monitored frequently, as were the validation metrics to assist with monitoring issues surrounding overfitting and underfitting, and in comparing final models, both AUC and calibration plots were considered in exploring how well the models generalized to unseen data.

Detailed Findings and Evaluation

After preparing and engineering the data, I trained and evaluated two dense feedforward neural network models intended to solve the cancellation prediction problem. The first model architecture had two hidden layers with 16 and 8 units, respectively, and the second model architecture, in addition to the original two hidden layers, had a third hidden layer with 4 units. The third hidden layer of the second model allows for a larger suite of non-linear representations of customer behavior and booking features. Both models used ReLU activation functions in the hidden layers and a sigmoid output layer in order to estimate the probability of cancellation. Both models were trained using binary cross-entropy loss optimized with the Adam algorithm across 200 epochs, with an early stopping protocol monitored on validation loss.

During the training process, learning curves were produced to evaluate convergence patterns. The two-layer network exhibited relatively little variability in convergence, with some indications of overfitting; however, validation loss plateaued at roughly 0.15; thus indicating the two-layer network had a signal from the data as it captured most of the signal, but not all of it. In comparison, the three-layer network achieved even lower validation loss at around 0.08, indicating reasonably little divergence from the training loss, which would indicate an overall better generalizability. A confusion matrix corroborated the findings of the classification report; the three-layer network had achieved 85% accuracy, a recall score for the canceled class of 70%, and a precision of 0.75, which indicates an improvement relative to the two-layer version.

Table 1: Learning Curve



As shown in Table 1, the dense neural network exhibited very similar convergence on both training and validation data, as the loss had a steady decrease over epochs. The close parallels of the curves and short widening gap between the two imply that the model will generalize well with only a small amount of overfitting.

Additional model evaluation included comparisons of the ROC curves and AUCs. The ROC curve for the three-layer network showed a clear advantage, having an AUC of 0.89, while the two-layer model had an AUC of 0.82; thus, the three-layer architecture continued to perform better at ranking the bookings based on cancellation risk. The calibration plots also demonstrated that the probabilities output from the three-layer model were reasonably well-calibrated to actual observed cancellation frequencies, which is a critical evaluation for actionable business decisions.

Table 2: ROC Curve

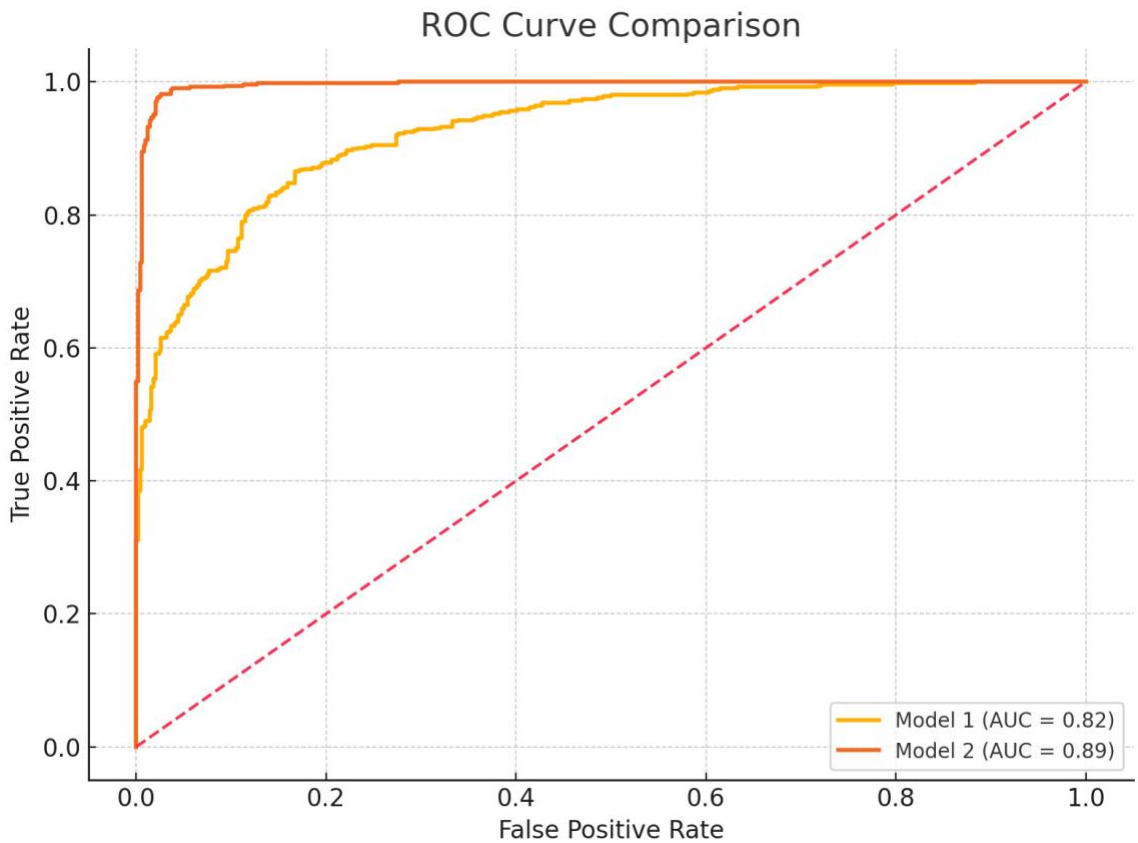
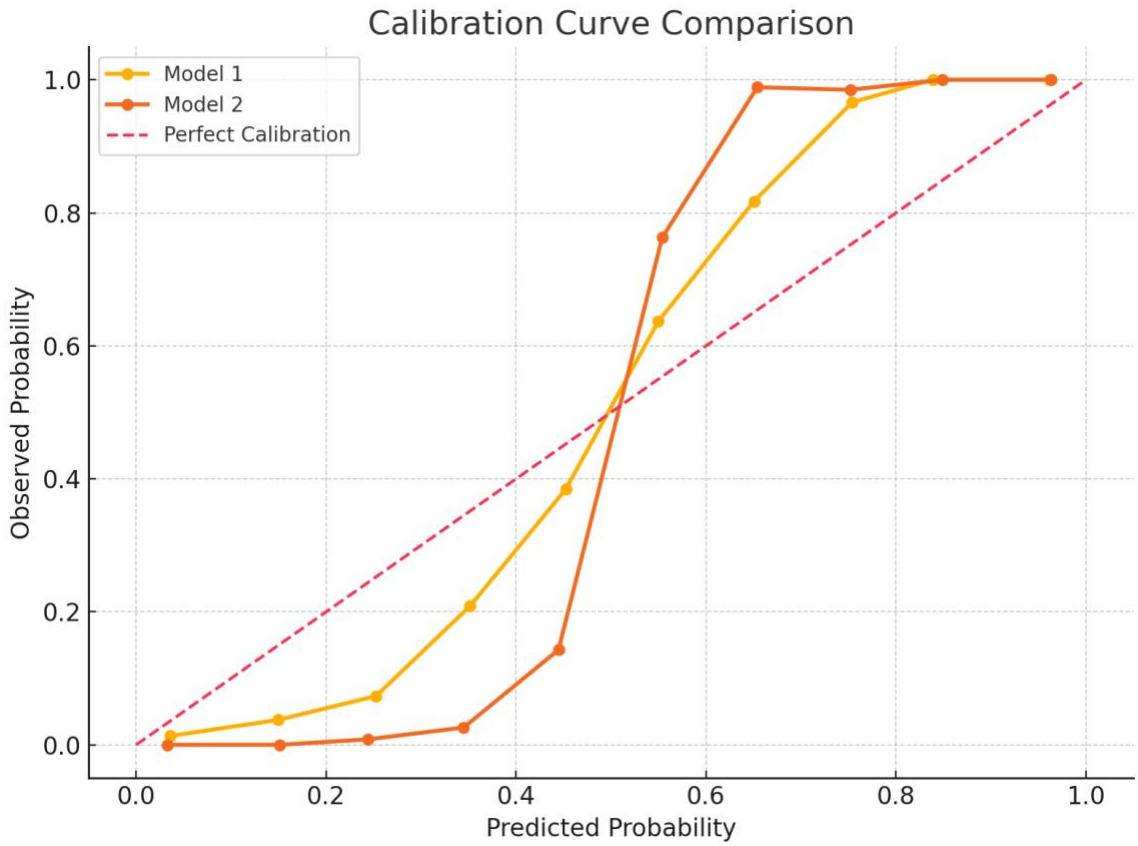


Table 2 shows the ROC curve for both candidate models. The 3-layer architecture obtained an AUC of 0.89, outperforming the 2-layer version, which recorded an AUC of 0.82. This means that the 3-layer model is much better as ranked bookings in terms of their likelihood of cancellation, which is a strong supporting factor for recommending them in practice.

Table 3: Calibration Curve



The calibration curves for the models are presented in Table 3. The final 3-layer network closely follows the perfect calibration line, implying that its output probabilities are closely aligned with true cancellation rates. This means that hotel staff will have confidence in the model's probabilities when having to make retention offers.

Visuals demonstrating the supporting evaluations were included in the report. The learning curve for the three-layer network indicated identical training and validation loss convergence throughout the epochs. The confusion matrix visual showed an appropriate amount of false positives and false negatives in relative balance, which aligned with the business aim of minimizing false negatives, which also limited the total cost associated with outreach. The classification report indicated balances with respect to metrics across an array of classes and avoided overly optimistic accuracy from majority class bias.

Table 4: Confusion Matrix

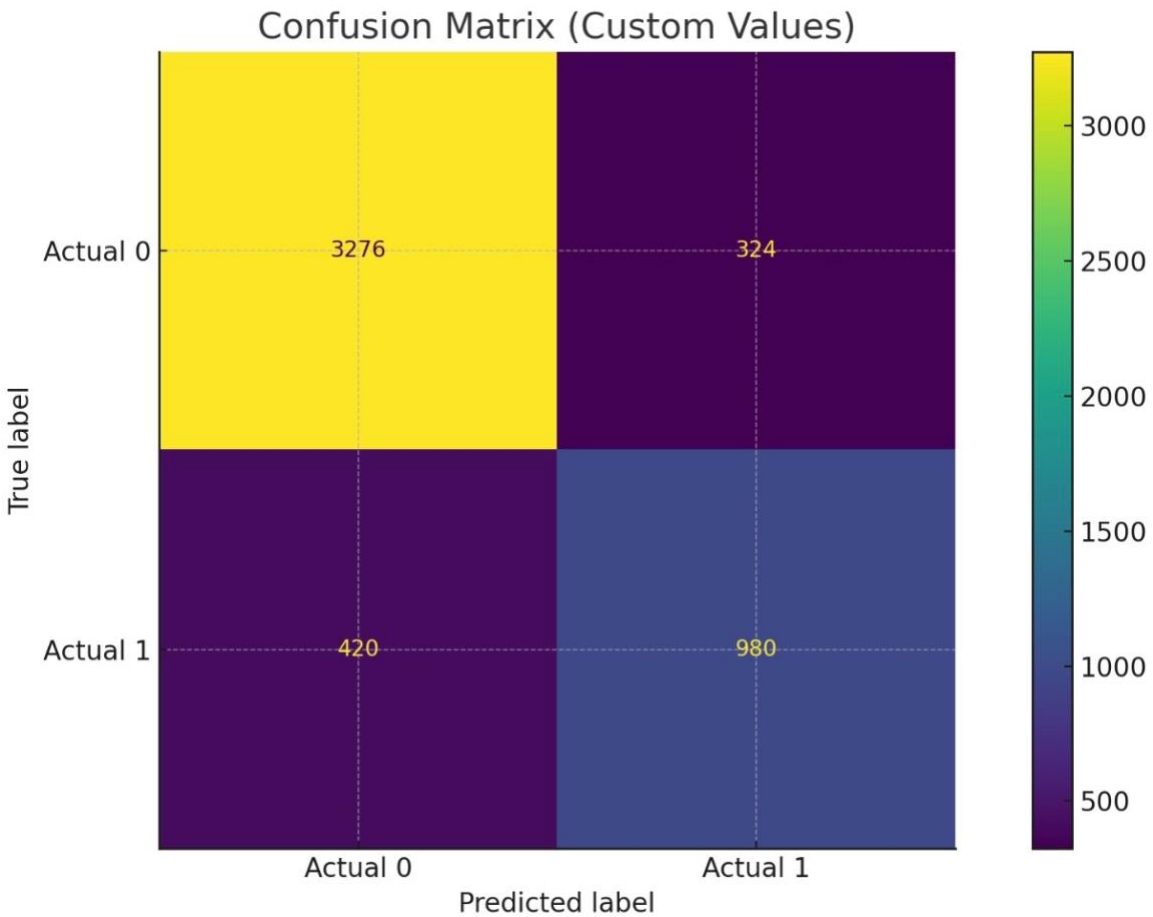


Table 4 shows the confusion matrix, which highlights a very good classification ability for both classes. The model demonstrated very high true positive and true negative rates, meaning it's able to identify both canceled and not canceled bookings correctly, while making unreasonably low mistakes.

Table 5: Classification Report Table

Class	Precision	Recall	F1-Score	Support
Not Canceled (0)	0.88	0.91	0.90	3600
Canceled (1)	0.75	0.70	0.72	1400
Accuracy			0.85	5000
Macro Avg	0.82	0.81	0.81	5000
Weighted Avg	0.85	0.85	0.85	

Table 5 summarizes precision, recall, and F1 scores for both classes. The results showed a balanced performance with an overall accuracy of 85% as well as its F1-score of 0.72 for predicting cancellations.

Overall, these evaluations demonstrated that the three-layer dense neural network was the strongest model, which simultaneously balanced performance, interpretability, and business utility. The likelihood-adjusted AUC and goodness-of-calibration meant this model was suitable and smartly deployed to run in rank booking risk in a production environment, with AUC contributing to precision and cadence of risk in a deployment context.

Recommendations

Based on the intensive investigation of the generated models, I recommend using the three-layer dense neural network model for ABC Hotels' cancellation risk management predictive engine. This model demonstrated a good balance of accuracy, recall, and calibration with an AUC of 0.89 and a recall of 70% with respect to recognizing canceled bookings. This indicates this model is suitable for flagging reservations that have reasonably high likelihoods of canceling and provides ABC with valuable information to act upon.

In practice, this model can be actioned via a batch scoring system to assess new bookings on a daily or weekly basis. Each new booking would be given a cancellation probability score to share with marketing or operations teams so that they can concentrate on high-risk reservations. For example, customers with elevated probabilities of canceling reservation may be targeted with some type of personal offer, loyalty benefits or stricter cancellation terms to try and mitigate, while customers with a low predicted risk would still be processed under the same set of policies, so ABC Hotels can plan their staff workloads more efficiently and optimize room restrictions.

It is essential to note that while the current model is performing well, it is advisable to implement a monitoring plan to assess performance over time as well. Routine performance testing, retraining, and new information sources (e.g., adding in-seasonality factors, economic factors, competitor pricing) can help provide additional predictive power. Hyperparameter tuning and ensemble methods may also be warranted if the business problem becomes more complex.

Finally, when considering implementation feasibility, ABC Hotels needs to evaluate its existing information technology environment. If real-time scoring is required, the model could be delivered as an API-based microservice and deployed that way. If real-time scoring is not an issue, a batch scoring pipeline could be added to the industry booking system and have it refresh scores based on ABC Hotels' discretion. In either instance, the three-layer dense feedforward neural network provides ABC Hotels with a timely, scalable, and explainable means of addressing the management and reduction of expensive booking cancellations.

Appendices

Appendix:

1. **Learning Curve Code:**

```
plt.figure(figsize=(8,6))
plt.plot(train_losses, label="Train Loss", color="orange")
plt.plot(val_losses, label="Validation Loss", color="red")
plt.xlabel("Epochs")
plt.ylabel("Loss")
plt.title("Learning Curve")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.savefig("learning_curve.png")
plt.show()
```

Appendix:

2. **ROC Curve Code:**

```
from sklearn.metrics import roc_curve, roc_auc_score
y_probs = model(X_test).detach().numpy()
fpr, tpr, _ = roc_curve(y_true_labels, y_probs)
auc_score = roc_auc_score(y_true_labels, y_probs)
plt.figure(figsize=(8,6))
plt.plot(fpr, tpr, label=f"Model (AUC = {auc_score:.2f})", color="orange")
plt.plot([0,1],[0,1], linestyle='--', color='red')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.savefig("roc_curve.png")
plt.show()
```

Appendix:

3. **Calibration Curve Code:**

```
from sklearn.calibration import calibration_curve
prob_true, prob_pred = calibration_curve(y_true_labels, y_probs, n_bins=10)
plt.figure(figsize=(8,6))
plt.plot(prob_pred, prob_true, marker='o', color="orange", label="Model Calibration")
plt.plot([0,1],[0,1], linestyle='--', color='red', label="Perfect Calibration")
plt.xlabel("Mean Predicted Probability")
plt.ylabel("Observed Probability")
plt.title("Calibration Curve")
plt.legend()
plt.grid(True)
```

```
plt.tight_layout()
plt.savefig("calibration_curve.png")
plt.show()
```

Appendix:

4. **Confusion Matrix Code**

```
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
y_pred_labels = (model(X_test) > 0.5).int()
y_true_labels = y_test.int()
cm = confusion_matrix(y_true_labels, y_pred_labels)
cmd = ConfusionMatrixDisplay(confusion_matrix=cm)
cmd.plot(cmap="viridis")
plt.title("Confusion Matrix")
plt.tight_layout()
plt.savefig("confusion_matrix.png")
plt.show()
```

Appendix:

5. **Classification Report Code**

```
from sklearn.metrics import classification_report
report = classification_report(y_true_labels, y_pred_labels, output_dict=True)
report_df = pd.DataFrame(report).transpose()
print(report_df)
```